

Trust the View That Sees the Target: Mining Cross-View Conflicts for Reliability-Gated Disaster Damage Assessment

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Abstract

After a disaster, building damage is assessed from overhead tiles and ground-level photographs, and most methods fuse the two views symmetrically, trusting both equally for every building. This paper focuses on the samples where that assumption fails: the *conflict cases*, on which two independently trained single-view models disagree. We mine such cases from three paired collections (inspection photographs from the 2025 Eaton wildfire and street-view panoramas from Hurricanes Ian and Milton, each matched to very-high-resolution overhead tiles), where they make up 10–33% of the data. On these samples an oracle that simply trusts the correct view beats every fusion method we tested by 0.37–0.41 accuracy, and the gap survives longer training, calibration, and backbone changes. We recover part of it with a visibility-conditioned reliability gate: a linear model that decides which view to trust from building-visibility features, calibrated per-view confidences, and the disagreement itself. On the wildfire data the gate is the only method that significantly beats calibrated probability averaging (+0.051 on conflicts, $p = 0.0001$) and end-to-end fusion (+0.072, $p < 10^{-4}$); on the panoramic datasets it matches them. A controlled field-of-view experiment explains why: cropping panoramas toward the building doubles the benefit of fusion, whereas random crops of the same size do not. Finally, the spatial density of conflicts predicts tile-level damage without labels (Spearman $r = 0.615$, $p = 0.001$). Mining conflicts turns “does fusion help?” into “which view should be trusted, where, and why?”

CCS Concepts

• **Computing methodologies** → **Computer vision**; *Machine learning*; • **Information systems** → *Geographic information systems*.

Keywords

cross-view fusion, street-view imagery, satellite imagery, disaster damage assessment, reliability estimation, GeoAI

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1 Introduction

Building-level damage assessment after a disaster increasingly draws on two large image collections at once. Overhead imagery covers a whole region within hours and is the basis of standardized damage mapping [2]. Ground-level imagery, whether crowdsourced street-view panoramas or official inspection photographs, shows the facades and debris that overhead sensors cannot see [10, 12]. Methods that pair the two views serve geolocalization and damage mapping [4, 5, 11], multimodal arbitration [9], and agentic assessment pipelines [7, 8]. Their usual conclusion is encouraging: fusion improves average accuracy over either view alone.

We argue that average accuracy hides both the real value and the real limitation of cross-view fusion. For most buildings the two views agree, and fusion changes little. The interesting records are the *conflicts*, the samples on which independently trained single-view models disagree. Conflicts concentrate where one sensor is blind: a facade can look intact from above and gutted from the street, or trees can hide a destroyed roof from the road. Finding these records in a large collection, deciding which view to believe for each, and reading their spatial pattern is as much a search-and-mining problem as a modeling problem.

We therefore make conflict cases the unit of analysis and ask three questions. First, how much information do fusion methods leave unused on conflicts? We measure an *oracle single-view gap*: the margin between an oracle that trusts whichever view is correct for each sample and the best available method. Across three disasters it is 0.37–0.41 accuracy, several times any fusion gain. Second, can the gap be recovered, and how? Per-sample view reliability turns out to be largely predictable from whether the ground image actually *sees the target building*, and a *linear gate* over visibility, calibrated confidence, and disagreement features recovers part of the gap while reading as a one-sentence rule. Third, is the effect causal? Datasets differ in hazard, geography, sensors, and labels at once, so we manipulate target alignment *within* a dataset with a controlled cropping experiment. A fourth result is directly useful for mining post-disaster collections: the spatial density of conflicts is a damage map that needs no labels (Figure 1). Code, protocols, and derived tables: <https://github.com/rayford295/CrossViewGate>.

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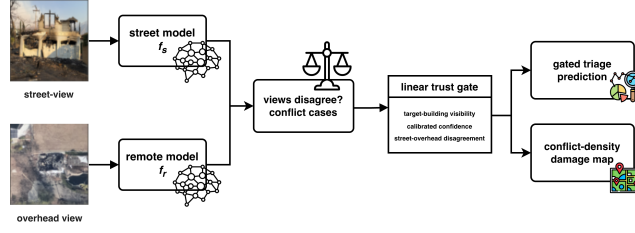


Figure 1: Overview. Independently trained single-view models define the conflict cases. A linear gate over target-building visibility, calibrated confidence, and cross-view disagreement decides which view to trust for each sample, and the spatial density of conflicts doubles as a label-free damage map. The pair shown is a real Eaton-fire conflict case.

Table 1: Test-set results (five-seed means; accuracy std ≤ 0.03 everywhere). Conflict accuracy is computed on the per-seed conflict subset; closure follows Eq. (1) with the oracle at 0.961 / 0.896 / 0.964 conflict accuracy (gap 0.410 / 0.367 / 0.407).

Method	Eaton wildfire (conflict rate 9.6%)			Hurricane Ian (33.2%)			Hurricane Milton (25.7%)		
	Acc.	Conflict acc.	Closure	Acc.	Conflict acc.	Closure	Acc.	Conflict acc.	Closure
street_only	0.931	0.486	—	0.703	0.529	—	0.759	0.557	—
remote_only	0.929	0.475	—	0.648	0.367	—	0.721	0.407	—
concat	0.930	0.674	0.29	0.709	0.588	0.15	0.766	0.622	0.16
crossview (end-to-end fusion)	0.940	0.699	0.35	0.735	0.624	0.25	0.776	0.649	0.22
calibrated probability averaging	0.954	0.732	0.44	0.731	0.617	0.24	0.776	0.626	0.17
reliability gate (linear)	0.959	0.768	0.52	0.731	0.618	0.23	0.777	0.631	0.18

2 Data and Protocol

We use three paired street/overhead collections spanning two hazards and, more importantly, two ground-view *observation regimes*: property-centric photographs aimed at a building, and panoramas that are not.

Eaton wildfire (Altadena, CA; property-centric). Building-level CAL FIRE Damage Inspection (DINS) photographs taken after the January 2025 fire, paired by location with post-event VHR overhead tiles. The six-level DINS scale is mapped to three ordinal classes (no/trace damage, damaged–repairable, destroyed). The split has 6.5k/1,966/1,984 train/val/test pairs. **Hurricane Ian (panoramic).** The CVIAN pairing released with CVDIsaster [5]: Mapillary 360° panoramas over Sanibel Island, 27 cm post-event satellite imagery, and three-level damage-perception labels, with 3,248/573/300 pairs. **Hurricane Milton (panoramic).** A 360° street-view/satellite pairing over Florida after the October 2024 hurricane, built in the same way; 1,740/307/254 pairs.

Splits and leakage. Eaton is split by property and tile, so all photographs of one structure fall in one partition. For Ian the headline numbers keep the released split, for comparability with CVDIsaster. However, an audit against the official CVIAN position file found that 299 of the 300 test samples share a Mapillary sequence with the training data, so we also evaluate a repaired protocol with 0.005° spatial blocks and a 25 m buffer. Under it, macro-F1 drops by 0.05–0.10 for every model: index-based splits overstate spatial generalization, although the single-view ranking (street above satellite) is stable.

3 Method

Base models and conflict cases. For each dataset we train four models under one protocol: *street_only* and *remote_only* (ResNet-18 [3] with a linear head), *concat* (both encoders with an MLP

head), and *crossview* (both encoders with a learned interaction head), which serves as the end-to-end fusion reference. Training uses class-balanced cross-entropy, 224-pixel inputs, AdamW, early stopping on validation macro-F1, and five seeds; a ResNet-50/CLIP/DINOv2 sweep changed no conclusion. Writing $s(x)$ and $r(x)$ for the two single-view predictions, the **conflict subset** is $C = \{x : s(x) \neq r(x)\}$. Conflict rates are 9.6% on Eaton, 33.2% on Ian, and 25.7% on Milton.

Oracle single-view gap. The oracle counts a conflict sample as correct if *either* single view is correct. For a method m we report its conflict accuracy $\text{acc}_C(m)$ and its **oracle-gap closure**,

$$\text{closure}(m) = \frac{\text{acc}_C(m) - \text{acc}_C(\text{best single})}{\text{acc}_C(\text{oracle}) - \text{acc}_C(\text{best single})}. \quad (1)$$

The oracle only chooses *which* of two fixed models to trust, so it bounds arbitration between existing predictors, not new models.

Calibration. Averaging miscalibrated probabilities can win or lose for reasons unrelated to information content [1]. We therefore fit one temperature per view on validation predictions and evaluate every training-free fusion baseline on calibrated probabilities; the argmax, and hence the conflict subset, is unchanged.

Visibility-conditioned reliability gate. A frozen SegFormer-B0 [6] trained on ADE20K gives four building-visibility features per street image: the building pixel ratio, the same ratio inside a centered crop of half the image size, their difference (a centering signal), and the normalized distance of the building centroid from the image center. Seven more features describe the predictions themselves: calibrated confidence and entropy for each view, the confidence gap, the Jensen–Shannon divergence between the two calibrated distributions, and a disagreement flag. From these eleven z-scored features the gate outputs softmax weights over the street, remote, and crossview probabilities, and is fit on the validation split by

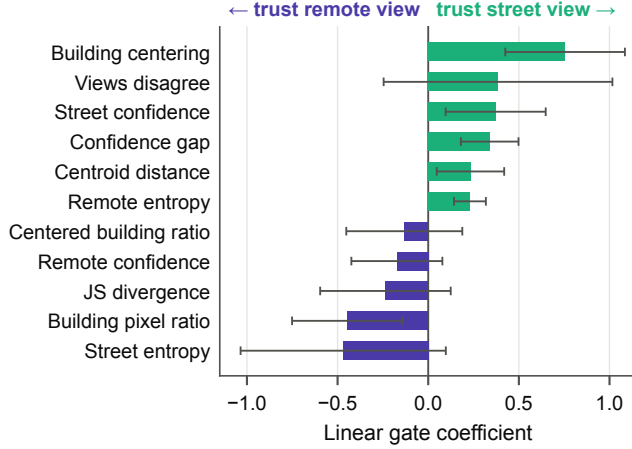


Figure 2: Linear gate coefficients on the wildfire (mean $\pm$ std over five seeds). Positive values shift trust toward the street view, negative values toward the remote view.

minimizing the negative log-likelihood of the gated mixture. The headline gate is *linear*: a small MLP matched it throughout, and linear coefficients can be read directly.

Field-of-view intervention. Each 1024×512 panorama is cropped to a 256×256 window with a 90° horizontal field of view on the horizon band. In the **building-centered** variant the window is aimed at the circular-mean azimuth of the SegFormer building mask. In the **random** variant it is aimed at a reproducible uniformly random azimuth. The two variants share size, projection, and information budget and differ only in whether the window points at the built structure. We retrain *street_only* and *crossview* on each variant (three seeds), reuse the remote models, and compare the gain of fusion on conflicts.

Conflict density. We aggregate seed-averaged predictions to geographic tiles and compute per tile the hard conflict density (fraction of samples on which the two argmax predictions disagree) and, as a control, the mean single-view predictive entropy. Both are correlated (Spearman) with the tile’s mean ordinal damage.

4 Results

Overall results and the oracle gap. All paired tests below pool each sample’s correctness difference over seeds and use a two-sided sign-flip permutation test (20,000 permutations). Table 1 shows three patterns. First, cross-view fusion beats both single views on conflicts on every dataset (crossview minus street: Eaton $+0.246$, $p < 10^{-4}$; Ian $+0.078$, $p = 0.013$; Milton $+0.141$, $p = 0.006$). Second, once the single-view models are converged and calibrated, simple probability averaging is a strong baseline: on Eaton it exceeds plain crossview (0.954 vs. 0.940), and end-to-end fusion does not significantly beat it anywhere (Eaton conflicts $+0.034$, $p = 0.073$). Reports that learned fusion beats simple aggregation may therefore reflect undertrained or miscalibrated baselines. Third, the oracle gap is much larger than any of these differences: knowing *which existing model to trust* for each sample would add 0.37–0.41 conflict accuracy. Calibration does not explain it, since calibrated averaging closes at most 44%

Table 2: Field-of-view intervention (three seeds; remote models unchanged). Conflict gain = crossview – best single view on conflicts.

Dataset	Street-view variant	Street acc.	Conflict gain	Closure
Ian	original panorama	0.709	+0.064	0.166
Ian	building-centered crop	0.648	+0.109	0.269
Ian	random crop (control)	0.574	+0.042	0.072
Milton	original panorama	0.740	+0.067	0.177
Milton	building-centered crop	0.693	+0.149	0.365
Milton	random crop (control)	0.654	+0.055	0.128

of the gap. The unused information concerns *when* each view is reliable.

The reliability gate. On the property-centric dataset the gate behaves as the mechanism predicts (Table 1). It gains $+0.072$ conflict accuracy over end-to-end fusion ($p < 10^{-4}$), $+0.018$ on the full test set ($p < 10^{-4}$), and $+0.051$ over calibrated averaging ($p = 0.0001$). Its closure of 52% is the only one above one half. On the panoramic datasets the street views rarely see the target (mean building pixel ratio 0.027 on Ian vs. 0.119 on Eaton), so the gate has little visibility signal to use and matches crossview (Ian $+0.005$, $p = 0.70$; Milton -0.015 , $p = 0.57$). This asymmetry is the central finding: gating pays where ground evidence is aimed at the target and costs nothing where it is not. The coefficients (Figure 2) keep their signs across seeds; street entropy has the largest negative weight, while building centering and the confidence gap are positive. In words: *trust the street view when it is confident and actually looking at the building*.

Field-of-view intervention. Building-centered cropping raises the conflict gain on both hurricanes (Table 2). On Milton the closure doubles, from 0.177 to 0.365, which is two-thirds of the wildfire level. Random cropping with identical geometry lowers the gain instead, and the building-minus-random contrast is positive in all six dataset–seed pairs. Building-centered cropping *lowers* street-only accuracy yet *raises* the benefit of fusion, so the gain comes not from a stronger street model but from the two views attending to the same object; the random control rules out cropping artifacts. Target alignment of the street view is therefore the causal variable behind the regime effect.

Conflict density as a spatial damage signal. On the wildfire, the conflict density of an overhead tile is a strong label-free predictor of its damage (Spearman $r = 0.615$, $p = 0.001$ over 25 tiles; $r = 0.60$ on a 0.01° grid; Figure 3). The uncertainty control behaves very differently: it *anti-correlates* with damage ($r = -0.37$ to -0.52), because parcels burned to the ground are easy, confident classifications. Conflict density is therefore not model uncertainty under another name but the spatial signature of two sensors disagreeing about the world. Damage on Eaton is spatially autocorrelated (Moran’s $I = 0.27$), so tile correlations partly reflect spatial structure. On Milton the signal drops below significance once the models converge ($r = 0.289$), because better single-view models disagree less; the wildfire result is the robust one.

5 Discussion and Conclusion

The 0.37–0.41 oracle gap measures the opportunity within a restricted choice between two existing single-view models, and it

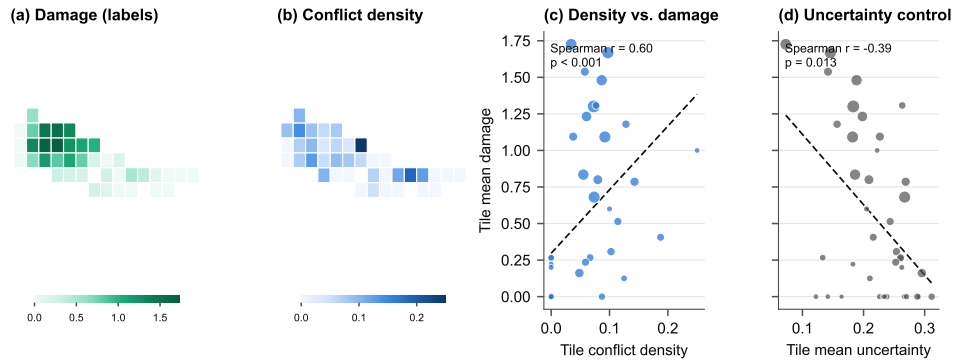


Figure 3: Conflict density as a label-free damage map (Eaton wildfire). (a) Labeled tile damage. (b) Tile conflict density computed without labels. (c) Their rank correlation. (d) The single-view uncertainty control, which anti-correlates with damage. Point size reflects tile sample count.

is far larger than any fusion gain. For mining paired geospatial collections, the productive question is therefore not “how should features be fused?” but “when should each view be believed?”. The gate closes half of the gap on property-centric imagery with eleven interpretable features; the remainder is a quantified target for richer visibility descriptors and pre-event reference imagery. In practice the gate adds one frozen segmentation pass per street image and a linear fit on existing output probabilities, and its rule fits in one sentence, which matters where automated triage must be auditable. Where ground imagery is panoramic, calibrated averaging captures most of the achievable benefit; where it is property-centric, as in inspection programs and insurance photography, gating pays. Because the driver is alignment rather than hazard, capture policy is itself an intervention: aiming cameras at structures raises the value of fusion.

Limitations. The property-centric regime rests on one wildfire dataset. Milton is small (2,301 pairs). The repaired CVIAN test set has only six spatial blocks, so strong spatial-generalization claims need more groups. The ADE20K segmenter measures built-structure visibility, not *the* target building. The density signal needs replication on further georeferenced disasters.

Conclusion. Where the two views disagree, symmetric cross-view fusion leaves 0.37–0.41 accuracy on the table on three real disasters. A linear, visibility-conditioned gate recovers a significant part of it, a field-of-view experiment shows that target alignment is the causal mechanism, and spatially aggregated disagreement is a label-free damage map. Mining conflicts turns “does fusion help?” into “which view should be trusted, where, and why?”.

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